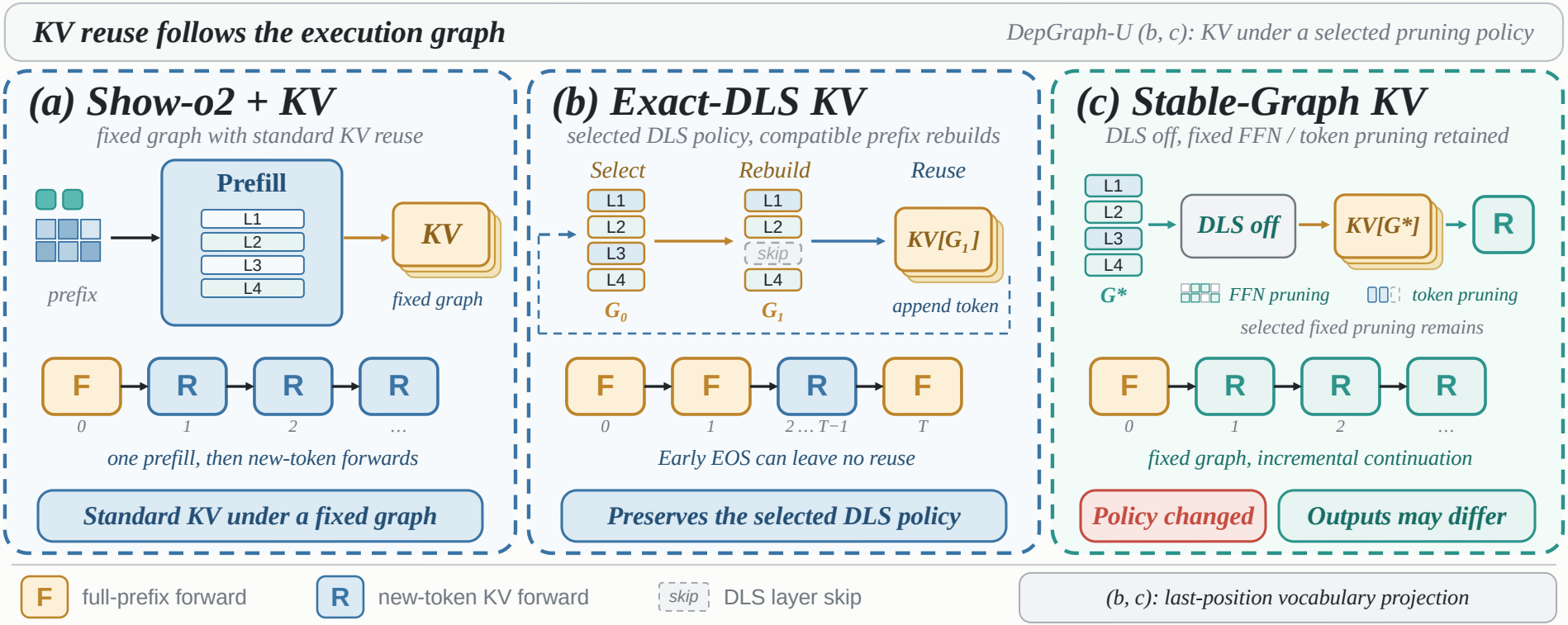

KV reuse follows the execution graph
DepGraph-U (b, c): KV under a selected pruning policy
(a) Show-o2 + KV
fixed graph with standard KV reuse
prefix
Prefill
L1
L2
L3
L4
KV
fixed graph
F
R
R
R
0
1
2
…
one prefill, then new-token forwards
Standard KV under a fixed graph
(b) Exact-DLS KV
selected DLS policy, compatible prefix rebuilds
Select
Rebuild
Reuse
L1
L2
L3
L4
L1
L2
skip
L4
KV[G1]
G0
G1
append token
F
F
R
F
0
1
2 … T−1
T
Early EOS can leave no reuse
Preserves the selected DLS policy
(c) Stable-Graph KV
DLS off, fixed FFN / token pruning retained
L1
L2
L3
L4
DLS off
KV[G*]
R
G*
FFN pruning
token pruning
selected fixed pruning remains
F
R
R
R
0
1
2
…
fixed graph, incremental continuation
Policy changed
Outputs may differ
F full-prefix forward
R new-token KV forward
skip DLS layer skip
(b, c): last-position vocabulary projection
(b, c) are alternative KV policies under pruning. F/R stop at EOS. T = DLS interval.